

MERCED 23 WSW, CA, USA

WMO: 745070

Lat: 37.2381N

Lon: 120.8825W

Elev: 78

StdP: 14.65

Time Zone: -8.00 (NAP)

Period: 04-19

WBAN: 93243

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	27.0	29.9	21.0	15.8	48.0	24.1	18.3	45.9	18.6	52.1	16.0	50.4	1.4	N/A	0.447

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	39.5	100.8	68.8	98.1	67.8	95.6	67.1	71.3	95.7	69.9	94.0	68.6	92.3	7.0	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.1	83.5	79.1	60.5	79.0	77.6	59.3	75.7	76.1	34.9	95.9	33.7	94.0	32.7	92.2	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.0	13.3	11.1	DB	21.9	105.2	2.9	2.0	19.8	106.7	18.1	107.8	16.5	108.9	14.4	110.3	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.2	46.1	49.9	54.4	58.2	64.3	70.9	75.3	73.5	69.8	62.0	52.4	45.6	(6)	
	DBStd	11.47	4.74	4.71	5.03	5.74	5.61	5.73	4.76	4.21	5.57	5.02	5.40	5.46	(7)	
	HDD50	413	139	55	17	4	0	0	0	0	0	0	37	161	(8)	
	HDD65	2772	587	422	329	217	82	12	0	1	20	120	379	603	(9)	
	CDD50	4152	17	53	153	249	443	627	784	729	593	373	108	23	(10)	
	CDD65	1039	0	0	1	12	60	190	320	265	163	28	0	0	(11)	
	CDH74	19887	0	0	77	498	1668	3834	5484	4557	2961	780	28	0	(12)	
	CDH80	10938	0	0	10	165	725	2167	3375	2681	1578	237	0	0	(13)	
(14) Wind	WSAvg	4.1	3.5	4.2	4.5	4.8	4.9	4.9	4.4	4.1	3.8	3.7	3.2	3.6	(14)	
(15) Precipitation	PrecAvg	10.7	2.5	2.2	1.4	0.6	0.6	0.1	0.0	0.0	0.1	0.6	0.9	1.8	(15)	
	PrecMax	21.5	6.3	8.5	4.4	2.3	3.7	0.3	0.2	0.2	0.5	2.5	2.8	3.6	(16)	
	PrecMin	5.2	0.3	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	(17)	
	PrecStd	3.8	1.6	1.7	1.2	0.6	0.8	0.1	0.0	0.0	0.1	0.7	0.7	1.1	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.9	72.0	81.0	91.0	97.3	103.6	104.5	102.2	100.0	92.0	77.5	65.7	(19)	
		MCWB	53.0	57.8	61.0	66.0	65.6	70.0	69.5	69.3	66.9	64.0	58.8	53.5	(20)	
	2%	DB	62.0	68.6	75.7	84.6	92.5	99.6	101.2	98.9	96.4	85.9	73.5	61.4	(21)	
		MCWB	52.5	56.6	59.3	62.0	63.2	69.0	68.5	68.1	66.4	62.8	58.7	52.2	(22)	
	5%	DB	59.2	65.0	72.3	80.0	87.9	95.7	98.3	95.9	93.3	82.0	69.4	58.7	(23)	
		MCWB	51.9	54.7	57.9	60.1	62.0	66.8	68.2	67.4	65.7	60.5	56.8	50.8	(24)	
	10%	DB	56.3	61.8	68.2	75.0	83.7	92.1	95.5	93.2	89.8	78.2	65.5	56.3	(25)	
		MCWB	51.2	53.6	56.2	58.5	61.2	65.7	67.8	67.1	64.8	59.5	54.4	49.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.3	62.0	63.5	68.1	68.6	74.0	72.9	72.0	71.0	67.1	63.4	60.1	(27)	
		MCDB	63.1	66.8	76.4	86.0	92.6	99.0	98.6	96.5	92.5	85.1	73.4	63.6	(28)	
	2%	WB	56.1	59.5	61.5	63.9	65.6	71.3	71.1	70.4	68.5	65.0	60.8	57.2	(29)	
		MCDB	59.9	66.1	72.6	81.1	88.2	95.8	96.3	94.2	90.3	82.2	69.6	59.4	(30)	
	5%	WB	54.4	57.7	59.6	61.8	63.7	69.2	69.7	69.0	67.3	63.3	58.3	54.4	(31)	
		MCDB	58.0	64.1	69.6	77.7	84.2	93.1	94.6	92.7	89.4	78.9	66.1	56.7	(32)	
	10%	WB	52.6	55.6	57.9	60.0	61.9	67.3	68.4	67.7	65.8	61.4	56.3	51.5	(33)	
		MCDB	56.1	61.6	67.2	73.6	81.4	90.7	92.9	90.9	87.9	75.3	63.6	55.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	19.3	23.1	25.3	30.0	34.6	38.8	39.5	39.4	37.5	31.5	25.9	20.5	(35)	
		MCDBR	25.6	29.2	32.7	38.3	41.8	43.1	41.2	41.6	41.6	37.7	31.4	23.5	(36)	
	5% WB	MCWBR	16.5	17.5	16.8	16.8	15.8	16.1	15.3	15.4	15.3	16.0	17.0	16.2	(37)	
		MCWBR	16.9	24.9	28.1	34.5	38.3	39.8	38.5	38.8	38.2	33.8	26.3	17.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.341	0.345	0.338	0.357	0.365	0.348	0.363	0.351	0.349	0.362	0.341	0.321	(40)	
		taud	2.421	2.427	2.447	2.394	2.385	2.446	2.414	2.426	2.439	2.401	2.438	2.507	(41)	
	Ebn at Noon		266	281	294	293	291	294	289	291	286	271	264	267	(42)	
		Edh at Noon	28	31	33	37	38	36	37	36	33	31	27	24	(43)	
(44) All-Sky Solar Radiation	RadAvg		761	1092	1556	1995	2382	2647	2578	2335	1924	1397	936	691	(44)	
	RadStd		112	123	140	164	123	86	72	61	59	79	55	98	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (46 neighbors)	+0.70	N/A	-2.09	N/A	N/A	N/A	N/A	-202	+252	+63	(47)

Nomenclature: See separate page